

LECTURE 7

Simon's algorithm:

Given $F: \{0,1\}^n \rightarrow \{0,1\}^m$ s.t. $\exists L \in \{0,1\}^n$

that satisfies

$$\forall x, y \in \{0,1\}^n \quad F(x) = F(y) \iff y = x \oplus L.$$

Goal: Find L .

$$\text{Note: } F(x) = F(x \oplus L) = F(x \oplus L \oplus L) = \dots$$

Recall

$$g: \{0,1\}^n \rightarrow \mathbb{C} \quad \text{with} \quad \frac{1}{2^n} \sum_{x \in \{0,1\}^n} |g(x)|^2 = 1$$

$$|g\rangle := \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} g(x) |x\rangle \quad (N=2^n)$$

$$= \frac{1}{2^{n/2}} \begin{bmatrix} g(0\dots 0) \\ g(0\dots 1) \\ \vdots \\ g(1\dots 1) \end{bmatrix}$$

$$\begin{aligned}
 |g\rangle \quad \begin{array}{c} - \boxed{H} \\ \vdots \\ - \boxed{H} \end{array} &\rightarrow g(0\dots 0) |\chi_{0\dots 0}\rangle + g(0\dots 1) |\chi_{0\dots 1}\rangle + \dots \\
 &= \hat{g}(0\dots 0) |0\dots 0\rangle + \hat{g}(0\dots 1) |0\dots 1\rangle + \dots
 \end{aligned}$$

$$\begin{aligned}
 |g\rangle &= g(0\dots 0) |0\dots 0\rangle + g(0\dots 1) |0\dots 1\rangle + \dots \\
 &= \hat{g}(0\dots 0) |\chi_{0\dots 0}\rangle + \hat{g}(0\dots 1) |\chi_{0\dots 1}\rangle + \dots \\
 \hat{g}(\vec{s}) &= \langle \chi_{\vec{s}} | g \rangle = \text{avg}_{\mathbf{z} \in \{0,1\}^n} \{ \chi_{\vec{s}}(\mathbf{z}) g(\mathbf{z}) \}
 \end{aligned}$$

Simon's algorithm:

Given $F: \{0,1\}^n \rightarrow \{0,1\}^m$ s.t. $\exists L \in \{0,1\}^n$

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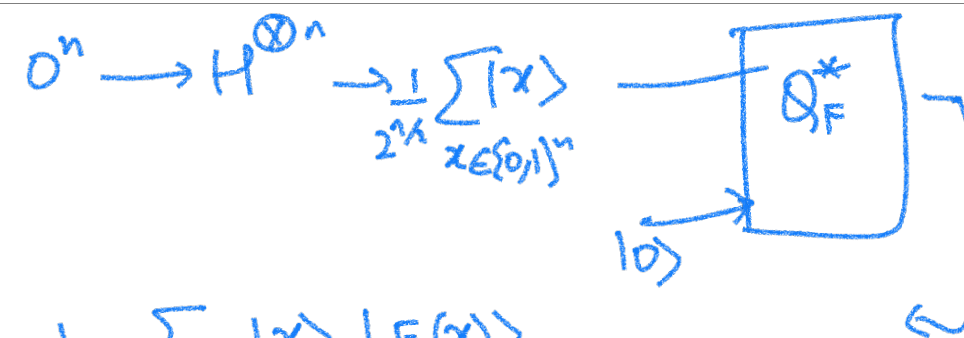
$$\forall x, y \in \{0,1\}^n \quad F(x) = F(y) \iff y = x \oplus L.$$

Goal: Find L .

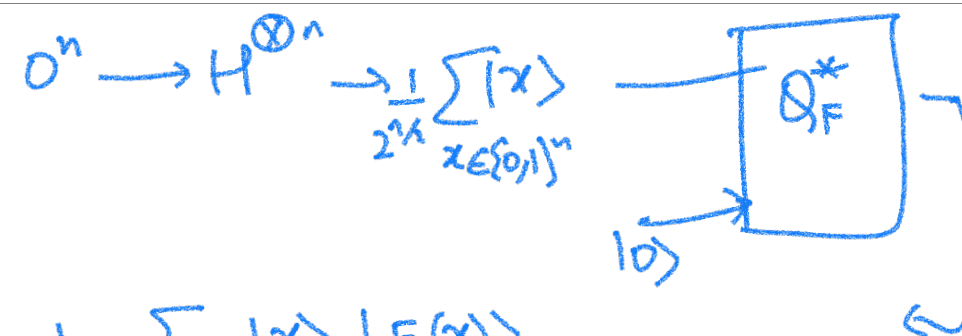
$$\text{Note: } F(x) = F(x \oplus L) = F(x \oplus L \oplus L) = \dots$$

Deterministic

Randomised



$$\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} |x\rangle |F(x)\rangle$$



$$\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} |x\rangle |F(x)\rangle$$

If we measure the last m qubits, we will see
one of $2^{m/2}$ outputs.

State collapses to $\frac{1}{\sqrt{2}} (|y\rangle + |y \oplus L\rangle) \otimes \dots$ for a random $y \in \{0,1\}^n$

$$\begin{array}{l}
 \text{if we repeat} \\
 |y_1\rangle + |y_1 \oplus L\rangle \\
 |y_2\rangle + |y_2 \oplus L\rangle \\
 \vdots
 \end{array}$$

Solution :

Step 1 : Create $\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |F(x)\rangle$, and measure last n qubits

State collapses to $\frac{1}{\sqrt{2}} (|y\rangle + |y \oplus L\rangle) \otimes |F(y)\rangle$

for some y .

Step 2: Hadamard on the 1st n qubits.

State changes to

$$\frac{1}{\sqrt{2}} \left(|x_y\rangle + |x_{y \oplus L}\rangle \right) \otimes |F(y)\rangle$$

↓

s^{th} co-ordinate

$$(-1)^{\sum y_i s_i \pmod{2}} + (-1)^{\sum y_i (L_i + s_i) \pmod{2}}$$

$$= (-1)^{\sum y_i s_i} \left[(-1)^{\sum L_i y_i} + 1 \right] = 0 \text{ if } \sum L_i y_i \text{ is odd.}$$

$$\frac{1}{2^{n/2}} \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ 1 \\ -1 \\ \vdots \\ 1 \end{bmatrix} \right)$$

Sign matches if
 $(-1)^{\langle y, s \rangle} = (-1)^{\langle y+L, s \rangle}$
 ie $(-1)^{\langle L, s \rangle} = 1$

Amplitude cancels
if $(-1)^{\langle y, s \rangle} \neq (-1)^{\langle y+L, s \rangle}$
i.e. $(-1)^{\langle L, s \rangle} = -1$

Step 3: Measure the 1st n qubits

We get a random y s.t. $\sum y_i L_i = 0 \pmod{2}$

We get $\vec{y}_1 \cdot \vec{L} = 0$

$$\vec{y}_2 \cdot \vec{L} = 0$$

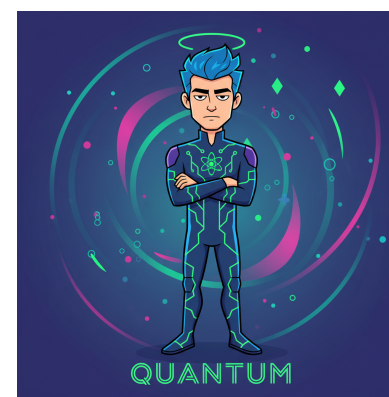
$\vdots$

Revisiting the Quantum Advantage



There is nothing
special about
Simon's algo

I can do what
you did!!

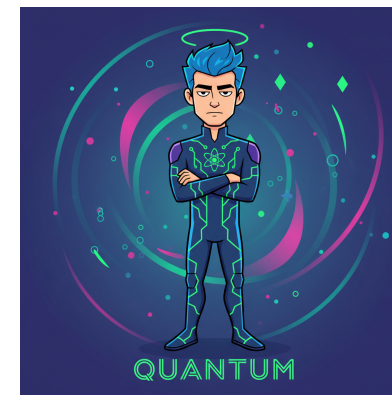


Really??



- Sample x at random from $\{0,1\}^n$
- Compute $f(x)$
- Output $(x, f(x))$

But I have all x in superposition

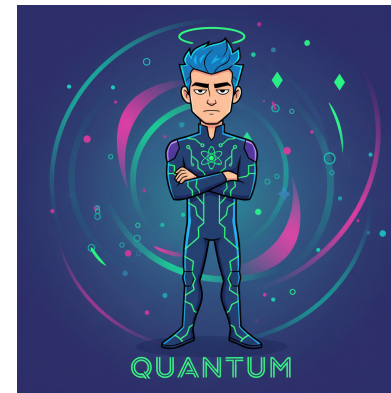




Oh! I will pretend to not look at x

Then?

Then, I will measure look at $f(x)$

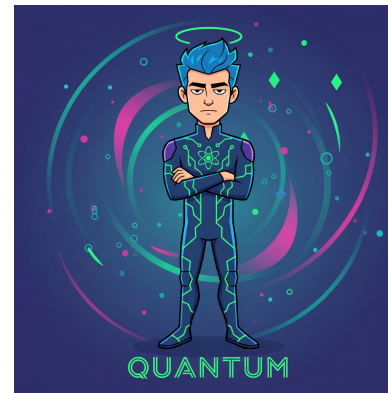




Now I have
w.p. $\frac{1}{2}$ $\vec{y}$
w.p. $\frac{1}{2}$ $\vec{y} + \vec{L}$
for a random $\vec{L}$

Now! I did
Fourier transf.
after this

Oh! That is
just coin tosses
with some -ve
signs thrown in!

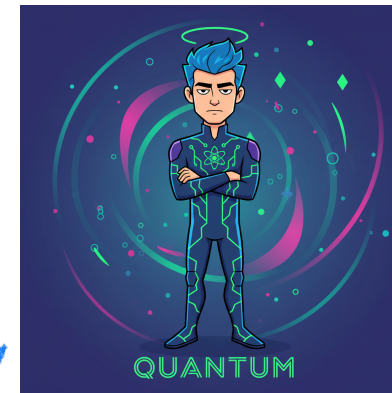




g got $\frac{1}{2^{1/2}} \begin{bmatrix} -1 \\ -1 \\ -1 \\ \vdots \end{bmatrix}$ w.p. $\frac{1}{2}$

and $\frac{1}{2^{1/2}} \begin{bmatrix} 1 \\ -1 \\ -1 \\ \vdots \end{bmatrix}$ w.p. $\frac{1}{2}$

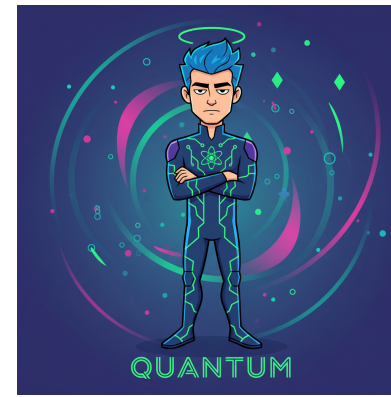
g got $\begin{bmatrix} -1 \\ -1 \\ -1 \\ \vdots \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ -1 \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \\ \vdots \end{bmatrix}$





Wait! What?

- How did probability vectors get added?
- How did probabilities cancel?





- Those are **not** probabilities!
- The word is **amplitude!**



Properties of the Fourier transform.

$$|x\rangle \xrightarrow{H^{\otimes n}} |\chi_x\rangle$$

$$|\chi_x(s)| = 1 \text{ for every } x, s$$

$$\chi_{x+y}(s) = \chi_x(s) \chi_y(s)$$

Helpful for the group structure $\mathbb{Z}_2^n = \{0,1\}^n$

$$\chi_{\vec{x} + \vec{x}}(\vec{s}) = \chi_{\vec{x}}(\vec{s})^2 = 1 \quad \forall \vec{x}, \vec{s}$$
$$= \chi_{\vec{0}}(\vec{s})$$

$$\text{Also } \langle \chi_x | \chi_y \rangle = \frac{1}{2^n} \sum_{\vec{s}} \chi_{x+y}(\vec{s}) = \begin{cases} 0 & \text{if } \vec{x} \neq \vec{y} \\ 1 & \text{if } \vec{x} = \vec{y} \end{cases}$$

We will now interpret $\{0,1\}^N$ as $\mathbb{Z}_N$, $N=2^n$,

$$g: \mathbb{Z}_N \rightarrow \mathbb{C}$$

We want to choose $\chi_0, \dots, \chi_{N-1}$

$$\chi_i: \mathbb{Z}_N \rightarrow \mathbb{C}$$

with similar properties

(This time, we are trying to figure out a useful transformation, and later see how to implement it)

Let's say we still want similar properties
(Remember, they helped us get Simon's algo)

$$|\chi_s(x)| = 1 \text{ for all } x, s.$$

$$\chi_s(x+y) = \chi_s(x) \chi_s(y)$$

$$\chi_s(\underbrace{x+x+\dots+x}_{N \text{ times}}) = \chi_s(x)^N = \chi_s(0) \quad \left(\begin{array}{l} \because \text{we are} \\ \text{in } \mathbb{Z}_N \end{array} \right)$$

$\chi_s(x)$ must be N -th root of unity.

Let $\omega_N = e^{2\pi i/N} = \cos \frac{2\pi}{N} + i \sin \frac{2\pi}{N}$

The N -th roots of unity

$1, \omega_N, \omega_N^2, \dots, \omega_N^{N-1}$

What should $X_S(x)$ be?



Natural choice

$$\chi_s(x) = \omega_N^{sx}$$

$$|s\rangle \xrightarrow{\boxed{U}} |\chi_s\rangle$$

Properties:

$$|\chi_s(x)| = 1 \quad \forall x, s$$

$$\begin{aligned} \chi_s(x+y) &= \omega_N^{s(x+y)} = \omega_N^{sx} \cdot \omega_N^{sy} \\ &= \chi_s(x) \chi_s(y) \end{aligned}$$

$$\langle x_s | x_t \rangle =$$

So $|x_0\rangle, \dots, |x_{N-1}\rangle$ are orthonormal

$$|s\rangle \xrightarrow{U} |x_s\rangle$$

$$U = \begin{bmatrix} |x_0\rangle & |x_1\rangle & \dots & |x_{N-1}\rangle \end{bmatrix}$$

Example: $N=8$, $\omega_8 = e^{2\pi i/8}$

$$U = \frac{1}{\sqrt{8}} \begin{bmatrix} \omega_8^0 & \omega_8^0 \\ \omega_8^1 & \omega_8^7 \\ \vdots & \vdots \\ \omega_8^7 & \omega_8^{49} \end{bmatrix}$$

$$g: \mathbb{Z}_N \rightarrow \mathbb{C} \quad \sum |g(n)|^2 = 1$$

$$|g\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} g(0) \\ g(1) \\ \vdots \\ g(N-1) \end{bmatrix} = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} g(i) |i\rangle$$

$$|g\rangle \xrightarrow{\text{DFT}_N} \sum_s \hat{g}(s) |s\rangle$$

"

$$\sum_s \hat{g}(s) |\chi_s\rangle$$

$$\begin{aligned} \hat{g}(s) &= \langle \chi_s | g \rangle \\ &= \frac{1}{N} \sum_z \chi_s^*(z) g(z) \\ &= \frac{\sum_z \omega_N^{-sz} g(z)}{N} \end{aligned}$$

$$\sum_s \hat{g}(s) |s\rangle \longrightarrow \boxed{\text{DFT}^{-1}} \longrightarrow \sum_s \hat{g}(s) |\chi_s\rangle$$

$$\text{DFT}^{-1} \quad |s\rangle \longrightarrow |\chi_s\rangle$$

$$\text{DFT} \quad |s\rangle \longrightarrow |\chi_s^*\rangle$$

DFT for $N=8$

$$\begin{bmatrix} \omega_8^0 & \omega_8^0 & \dots & \omega_8^0 \\ \omega_8^{-1} & \omega_8^{-2} & & \omega_8^{-7} \\ \vdots & \vdots & & \vdots \\ \omega_8^{-7} & \omega_8^{-14} & & \omega_8^{-49} \end{bmatrix}$$

Simon's algorithm for $\mathbb{Z}_N$

$$f: \mathbb{Z}_N \rightarrow \mathbb{C}$$

Period: L

$$f(x) = f(x+L) = \dots$$

$L|N$ we need to find L .

(This is an artificial setting.

We will later see how to use it for an actual
algo to factor integers)

$$|0\rangle \xrightarrow{\text{DFT}_N} \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle \xrightarrow{Q_F}$$

How to implement DFT.

Consider $N = 16$.

$$|x\rangle \xrightarrow{\text{DFT}_{16}} \frac{1}{\sqrt{16}} \sum_{s=0}^{15} \chi_s(x)^* |s\rangle$$
$$= \frac{1}{\sqrt{16}} \sum_{s=0}^{15} \omega_{16}^{-sx} |s\rangle$$

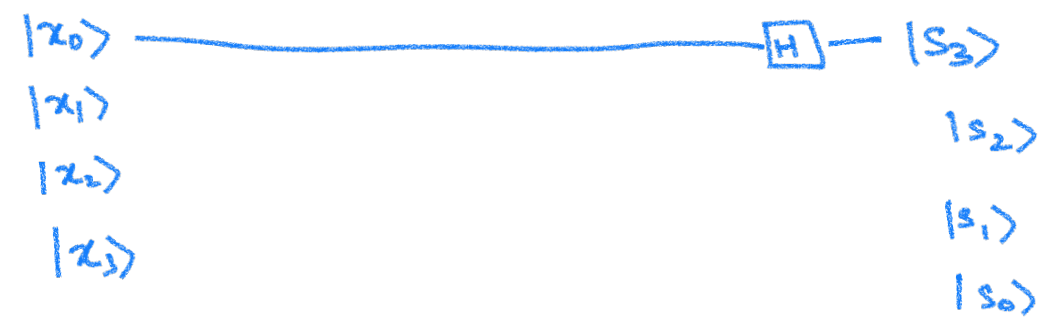
$$= \frac{1}{4} \left(|0000\rangle + \omega^{-x} |0001\rangle + \dots + \omega^{15x} |1111\rangle \right)$$

Unentangled form?

$$\left(\frac{|0\rangle + \bar{\omega}^{8x} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \bar{\omega}^{4x} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \bar{\omega}^{2x} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \bar{\omega}^x |1\rangle}{\sqrt{2}} \right)$$

Challenge : Each output qubit depends on the whole of x .

Question : Does $\frac{|0\rangle + \bar{\omega}^{8x} |1\rangle}{\sqrt{2}}$ depend on all of x_0, x_1, x_2, x_3 ?



lets look at $|s_2\rangle$ now

$$\text{Need } \frac{|0\rangle + \bar{w}^4 |1\rangle}{\sqrt{2}}$$

$$= \frac{|0\rangle + \bar{w}^{4(x_0 + 2x_1 + 4x_2 + 8x_3)}}{\sqrt{2}} =$$

$$|s_2\rangle = \frac{|0\rangle + \bar{\omega}^{4x_0} \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}}$$

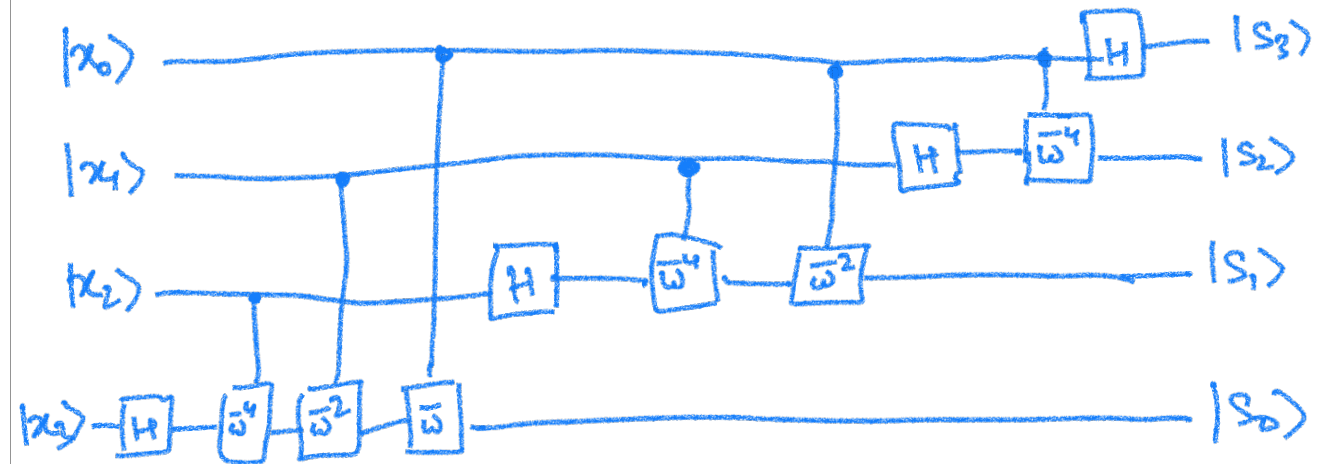
$$|x_1\rangle \xrightarrow{H} \frac{|0\rangle + (-1)^{x_1} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}}$$

$$|x_0\rangle \xrightarrow{\text{CNOT}} \frac{|0\rangle + \bar{\omega}^{8x_1} |1\rangle}{\sqrt{2}} \xrightarrow{\bar{\omega}^4} |s_2\rangle$$

$$|s_1\rangle = \frac{|0\rangle + \bar{\omega}^{2x} |1\rangle}{\sqrt{2}} = \frac{|0\rangle + \bar{\omega}^{2x_0 + 4x_1 + 8x_2} |1\rangle}{\sqrt{2}}$$

$$|x_2\rangle \rightarrow \boxed{H} \rightarrow \frac{|0\rangle + (-1)^{x_2} |1\rangle}{\sqrt{2}}$$

Final circuit



No. of gates $1+2+3+\dots+n = \frac{n(n+1)}{2}$